europass

Syed Hassan Zaheer

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ABOUT ME

Computer Science student at the University of Agriculture Faisalabad with experience in **Machine Learning, Flutter development, MERN Stack, Python, and Linux-based systems**. Skilled in building **scalable applications and AI-driven solutions**, with additional experience in **IoT development**. Passionate about solving complex problems and delivering **efficient, reliable, and user-focused software solutions**.

EDUCATION AND TRAINING

2023 – CURRENT
BS COMPUTER SCIENCE University of Agriculture, Faisalabad

Website https://web.uaf.edu.pk/

2021 – 2023 Faisalabad, Pakistan
INTERMEDIATE(FSC) Kips college, Sargodha Road

Final grade A+

2019 – 2021 Faisalabad, Pakistan
MATRIC Zeenat Modern Public High School

Final grade A+

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PROJECTS

Handwriting Recognition System — CNN Deep Learning

Technologies: Python, TensorFlow, Keras, Google Colab, EMNIST Dataset

- Developed an **offline desktop application** to recognize **handwritten letters and digits** using a **Convolutional Neural Network (CNN)**.
- Trained the model on the **EMNIST dataset (800,000+ images)** covering **62 classes (A–Z, a–z, 0–9)**.
- Designed a **CNN architecture with 3 convolutional layers**, **Batch Normalization**, **Dropout**, and **Fully Connected layers** to improve generalization.

- Built a **real-time drawing interface (280×280 canvas)** allowing users to write characters and receive **instant AI predictions**.
- Implemented an **image preprocessing pipeline** including **grayscale conversion, segmentation, padding, resizing to 28×28, and normalization**.
- Achieved **88.12% test accuracy** after **50 training epochs using Google Colab GPU**.

Link https://github.com/SyedHassan904/Convolutional_neural_network

IoT-Based Weather & Air Quality Monitoring System

Technologies: ESP32, DHT11, MQ135, C++, React.js, Firebase, REST APIs

- Built an IoT system capable of measuring **temperature, humidity, and air quality (AQI)** using ESP32 with DHT11 and MQ135 sensors.
- Programmed the ESP32 in **C++** to collect real-time sensor data and transmit it efficiently to the backend server.
- Integrated **Firebase** for real-time data storage, synchronization, and remote access to sensor readings.
- Created a responsive **React.js dashboard** to visualize live environmental data with charts, notifications, and real-time updates.
- Designed the system for **continuous environmental monitoring**, enabling precise and real-time AQI reporting.

Link https://github.com/SyedHassan904/dht-mq135-aqi-app.git

Image Stock Website — MERN Stack

Technologies: MongoDB, Express.js, React.js, Node.js, JWT, Cloud Storage, Vercel

- Developed a full-stack **image stock platform** using the MERN stack with complete user and image management workflows.
- Implemented **authentication and authorization** using JWT and secure role-based access control.
- Built features for **password reset, image upload, image selling, and buying**.
- Designed a responsive and clean **React UI** with optimized API routes for fast loading and smooth navigation.
- Integrated backend services for real-time image handling and database operations.
- Successfully **deployed the application on Vercel**, ensuring high availability and performance.

Link https://visualstock.vercel.app/

C# Windows Projects

1. **imgEditor – Grayscale Image Converter**
   - **Technologies:** C#, Windows Forms, .NET Framework
   - Developed a **desktop application** to convert colored images to grayscale using **pixel-by-pixel manipulation**.
2. **Pharmacy Management System**
   - **Technologies:** C#, Windows Forms, .NET Framework, MS Access
   - Built a **Windows Forms application** to manage pharmacy operations, including **admin login, pharmacist account creation, and secure access control**.
   - Integrated **MS Access (PharmacyDB.accdb)** to handle user accounts and data securely.

Link https://github.com/SyedHassan904/Visual_Programming_CSharp.git

TECHNICAL SKILLS

Skills

- **Programming Languages:** Python, C++, C#, Java, JavaScript, Dart, Assembly Language
- **Machine Learning & Data Science:** TensorFlow, Keras, Convolutional Neural Networks (CNN), Pandas, NumPy, Data Visualization, Predictive Modeling
- **Frontend Development:** React.js, HTML, CSS, JavaScript, Flutter
- **Backend Development:** Node.js, Express.js, REST APIs
- **Databases:** MongoDB, MySQL, Firebase

- **IoT & Embedded Systems:** ESP32, DHT11, MQ135, Sensor Data Processing
- **Version Control:** Git, GitHub
- **Operating Systems & Tools:** Linux, Windows, Visual Studio, VS Code, Google Colab

HOBBIES

Hobbies and interests

- Problem solving and logical puzzles
- Reading tech blogs and tutorials
- Learning and practicing programming